Satyam Patel

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SUMMARY

AI Engineer who designs and ships LLM-powered workflows end to end. At Interact Health PRO, I lead a 10+ member cross-functional team automating healthcare intake, onboarding, and reporting with Claude, Gemini, Airtable, and Make.com. Previously built a RAG chatbot for a legal-tech startup that cut LLM hallucinations by 98% with sub-1.5 s retrieval across 25k documents, and improved pose-estimation accuracy by 18% for a CCTV computer-vision startup. Grounded in reinforcement learning (University of Alberta) and applied computer-vision research.

EDUCATION

University of Alberta **Apr 2026**
Specialization in Reinforcement Learning *Edmonton, AB*

Conestoga College **Dec 2025**
Post Graduate Program, Big Data Solution Architecture *Kitchener, ON*

Conestoga College **Aug 2024**
Post Graduate Program, Artificial Intelligence and Machine Learning *Waterloo, ON*

Lovely Professional University **Dec 2020**
Bachelor of Technology, Computer Science *India*

EXPERIENCE

Interact Health PRO (IHP)
Project Manager – AI Automation Engineer *Kitchener, ON* · **May 2026 – Present**

- Lead project coordination for a 10+ member cross-functional intern team, using Scikit-Learn and Google Gemini to design AI-supported healthcare rehabilitation coordination, client intake, and medical-legal workflow prototypes, resulting in faster intake processing and reduced manual effort
- Architect structured Claude AI workflows with Claude Skills, Routines, and CLAUDE.md, creating repeatable SOPs that replace manual administrative tasks across intake, onboarding, and reporting, which improved process consistency and cut task completion time
- Build automation infrastructure using Airtable and Make.com, engineering conditional-logic triggers and task-based AI agents to route communications, track progress, and generate operational reports, enabling real-time visibility and decreasing reporting latency
- Direct sprint planning and workflow-mapping sessions with Operations, Development, and Analyst teams, maintaining process documentation in Git and conducting testing cycles, which ensured automation reliability and on-time delivery of features

Legal-Tech Startup *(contract-related tax guidance)*
RAG Chatbot Developer *Remote* · **Jan 2026 – Aug 2026**

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) chatbot using LangChain and PyTorch on AWS to answer users' contract-related tax questions, applying Test-Driven Development and Git version control, which cut LLM hallucinations by 98% and produced factual, context-aware outputs
- Integrated the Pinecone vector database containing over 25,000 embedded documents, enabling rapid semantic search and retrieving precise domain knowledge in under 1.5 seconds
- Fine-tuned the base language model on proprietary data and deployed the chatbot into the startup's production application, which drove a 35% increase in user engagement

Computer Vision Startup *(CCTV human detection & movement tracking)*
Computer Vision Data Specialist *Remote* · **Dec 2025 – Feb 2026**

- Boosted human pose estimation accuracy by 18% in real-world environments by identifying and resolving over 1,000 complex occlusion edge cases within the training dataset
- Refined spatial alignment for ground-truth data across 10,000+ images by recalibrating overlapping body keypoints with in-house script tools, which improved input fidelity for the computer-vision pipeline
- Decreased false-positive detection rates by 22% through rigorous data quality control and manual keypoint annotation, directly solving the model's inability to handle overlapping limbs

Conestoga College
Graduate Research Assistant *Waterloo, ON* · **Jan 2024 – Aug 2024**

- Engineered "The Plant Guard" system and conducted academic research to address precision-agriculture challenges by designing end-to-end computer-vision pipelines with TensorFlow and OpenCV, enabling early detection of plant diseases and reducing manual scouting time
- Processed and augmented a dataset of tens of thousands of images using Python and Pandas, then delivered the cleaned data to leadership so they could make informed decisions
- Architected and trained multiple neural networks using TensorFlow and Keras, establishing a baseline with a custom Vanilla CNN and improving feature extraction by implementing Transfer Learning with a fine-tuned VGG16 model
- Achieved 90%+ classification accuracy across multiple plant species and pathology categories, delivering a scalable, modular machine learning pipeline optimized for future deployment on mobile and edge computing devices

PROJECTS

Autonomous LLM Reasoning Agent (RLVR / GRPO) PyTorch, GRPO, LoRA, Unsloth, Hugging Face

- Aligned a Qwen model for autonomous chain-of-thought reasoning with GRPO (RLVR), replacing the critic and reward models with verifiable regex-based rewards
- Ran the full multi-generation RL training loop on a single 16 GB T4 GPU using Unsloth, LoRA, and 4-bit quantization, avoiding multi-GPU A100 clusters
- Built deterministic experimentation frameworks to evaluate model outputs reproducibly
- Published the trained model on Hugging Face

Multi-Agent Text-to-SQL Data Analyst LangGraph, Llama-3.3 (Groq), MongoDB, Docker

- Migrated a multi-agent Generative AI application to a self-hosted **n8n** server, automating complex **LangGraph** LLM workflows to translate natural language into SQL, which reduced manual query processing time by 85%
- Re-architected the backend infrastructure by transitioning to a highly scalable **MongoDB** database, optimizing the system to handle enterprise datasets and supporting 50,000+ concurrent real-time requests with sub-second latency
- Productionized real-time AI microservices using **FastAPI** and **Docker**, engineering automated CI/CD pipelines that accelerated model deployment cycles by 40% while ensuring robust data governance and security
- Designed a self-correcting agent loop: a SQL Writer (Llama-3.3-70B), a sandboxed Executor/Critic that reads its own tracebacks and retries, and an Analyst agent that turns query results into business insights

Autonomous Cloud Resource Optimizer Stable-Baselines3 (PPO), Gymnasium, Streamlit, TensorBoard, Plotly

- Engineered a custom Reinforcement Learning environment using Python and Gymnasium to simulate network traffic, and trained a PPO agent with Stable-Baselines3 to autonomously scale cloud servers up or down based on demand
- Designed a specialized reward function that balanced system performance with power efficiency, successfully reducing simulated cloud infrastructure costs by 30% while maintaining reliable server uptime
- Packaged the trained model into a Dockerized FastAPI microservice with RESTful APIs, creating a ready-to-deploy CI/CD solution that can easily integrate with platforms like Microsoft Azure and AWS for real-time resource management

AI Automation Pipeline Suite *n8n, Google Cloud Vertex AI, Veo, Gemini, Claude API, FFmpeg*

- Built an end-to-end AI video production pipeline in n8n that converts a single text prompt into a 12-scene storyboard, AI-generated video clips, captions, narration assets, and render-ready outputs using Gemini 2.5 Pro and Veo
- Architected a resilient asynchronous orchestration layer for long-running generation jobs with polling, retry logic, cloud status tracking, GCS URI-to-HTTPS conversion, and automated asset handling, keeping per-clip delivery under 3 minutes
- Automated post-production and publishing by generating SRT caption files, FFmpeg concat manifests, structured output paths, quality-filtered downloads, and YouTube uploads with metadata, privacy settings, and video tracking handled inside n8n
- Developed a multi-brand AI product feed optimization system using LLM APIs to process full product catalogs and generate Google Merchant Center titles, Google Ads copy, Meta ad assets, and campaign-ready marketing files from a centralized brand configuration

CERTIFICATIONS

- Microsoft AI & ML Engineering: Microsoft Azure for AI and Machine Learning
- Microsoft Building Intelligent Troubleshooting Agents: Microsoft Azure AI: Workloads and Machine Learning on Azure
- Microsoft Professional Program in Artificial Intelligence
- TCPS 2: CORE
- COC Workshop
- Amazon Alexa Developer

TECHNICAL SKILLS

- **Generative AI & LLMs**: LangChain, LangGraph, Vector Databases, RAG Systems, OpenAI APIs, Open-Source LLMs (DeepSeek, Llama-3.3), Prompt Engineering, LoRA, Hugging Face, Foundational Models, Agentic AI, Google Gemini, Vertex AI, Veo, Text-to-Speech (TTS)
- **Machine Learning & Deep Learning**: PyTorch, TensorFlow, Keras, Scikit-Learn, NLP, Feature Engineering, Statistical Modeling, Anomaly Detection, Data Mining, Model Drift Monitoring, Deep Reinforcement Learning
- **Reinforcement Learning**: PPO, GRPO, Stable-Baselines3, Custom MDP Design, Power Efficiency Modeling, Gymnasium, Custom SARSA
- **Data Engineering & Languages**: Python (NumPy, Pandas, Scikit-learn), SQL (PostgreSQL, SQLite), MongoDB (NoSQL), C/C++, JavaScript, Apache Spark, Hadoop, Data Pipelines, Data Structures & Algorithms
- **MLOps & Cloud**: Docker, FastAPI (Microservices), RESTful APIs, CI/CD Pipelines, Microsoft Azure, AWS, Streamlit, TensorBoard, Git, Linux, DevOps, Test-Driven Development (TDD), Google Cloud Platform (GCP), Google Cloud Storage (GCS)
- **XAI & Interpretability**: Visualization Techniques, Reward/Feature Attribution, SHAP, LIME
- **AI Orchestration & Automation**: Claude (Skills, Routines, CLAUDE.md), n8n, Make.com, Airtable, Multi-Agent Workflows, Semantic Routing / Tool Calling, Apache Kafka / RabbitMQ, ETL / ELT Automation, Webhooks & API Integration, YouTube Data API v3, RapidAPI, OAuth 2.0 Integration
- **Languages**: English and Hindi (bilingual), French (intermediate)